TERMS OF REFERENCE (TOR)

Development & Deployment of an On-Demand Multi-Merchant Delivery Application Ecosystem

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1. PROJECT OVERVIEW & OBJECTIVES

1.1 Introduction

This Terms of Reference (TOR) defines the functional, technical, and operational requirements for contracting a qualified software development partner to design, develop, test, and deploy a robust, highly-scalable, and secure on-demand multi-merchant delivery ecosystem. The ecosystem will consist of cross-platform mobile apps for customers and drivers, coupled with management portals for merchant partners and system administrators.

1.2 Project Objectives

The software development firm is expected to deliver a high-performance system aimed at accomplishing the following core goals:

- **Seamless Automation:** Fully automate order placement, merchant dispatch, driver assignment, and routing optimized by real-time geolocation engines.
- **Market Competitive UX:** Provide intuitive, low-latency interfaces comparable to leading global delivery utilities, ensuring customer retention and high conversion rates.
- **Data Integrity & Transparency:** Establish immutable transaction records, multi-party electronic ledger accounting, and live end-to-end operational visibility.

1.3 Target Audience

The system directly services three user domains: consumer end-users seeking on-demand fulfillment, localized independent and enterprise merchants seeking digital dispatch extension, and fleet couriers requiring flexible task assignment engines.

2. SCOPE OF WORK

The selected vendor will execute end-to-end product realization, encompassing UI/UX prototyping, multi-tier software architecture engineering, native/cross-platform compilation, API synthesis, cloud DevOps configuration, QA staging, and production deployment.

2.1 Module 1: Customer Application (iOS & Android)

- **Identity & Profile:** OAuth2 social sign-ins, phone-number OTP validation via SMS, and profile profile/vaulted multi-address books.
- **Discovery Engine:** Smart geolocation filtering, category indexing, global search with fuzzy logic auto-suggest, and dynamic promotional banners.
- **Checkout & Settlement:** Real-time cart calculations with item variables, promotional coupon validation, and multi-gateway secure card processing (Stripe, digital wallets, Cash on Delivery).
- **Live Telemetry:** Interactive map interface powered by background web-sockets for absolute order phase tracking from merchant preparation to courier doorstep.

2.2 Module 2: Driver / Rider Application (iOS & Android)

- **Onboarding & Duty Toggle:** Independent courier background verification log uploads and a simple 'Online/Offline' active duty switch.
- **Intelligent Dispatch:** Interactive push notifications detailing mileage, guaranteed payouts, and strict timeout countdowns for fair job accept/reject flow.
- **Turn-by-Turn Telemetry:** Deep integration with mapping SDKs providing optimal route matrix routing across pickup and delivery nodes.
- **Proof of Delivery (PoD):** In-app optical character recognition (OCR) barcodes scanning, signature capture pad, and timestamped multi-angle photographic upload.

2.3 Module 3: Merchant Dashboard (Responsive Web Panel)

- **Order Lifecycle Management:** Sound-alert order intake console, manual preparation latency selection, and immediate rider dispatch triggers.
- **Dynamic Catalog Matrix:** Real-time inventory status toggle, dynamic price shifts, and configurable item modifiers/addon management.
- **Financial Settlement:** Consolidated automated statement logs tracking partner commissions, net payouts, and historical transaction exports.

2.4 Module 4: Super Admin Control Core (Web Console)

- **Global Control Tower:** Interactive map cluster tracking every active transaction state, driver coordination vector, and regional order density metrics.
- **Financial Configuration:** System-wide base fee matrices, dynamic distance pricing modifiers, surge calculation configuration, and commission splits.
- **User Authorization:** Granular Role-Based Access Control (RBAC) separating customer service agents, fraud auditors, and database engineers.

3. TECHNICAL REQUIREMENTS

The software vendor must guarantee adherence to modern architectural patterns. Microservices framework deployment via cloud container instances is highly prioritized.

3.1 Recommended Technology Stack Blueprint

Component	Preferred Architecture / Stack Options
Mobile Applications	Flutter SDK or React Native (Cross-Platform compiled for iOS/Android using single codebase optimization) or fully Native Swift/Kotlin.
Backend Core & APIs	Node.js (TypeScript), Go (Golang), or Python (FastAPI). RESTful and WebSocket protocols for low latency data synchronization.
Primary Database Core	PostgreSQL or MySQL (Relational transactional data); MongoDB or DynamoDB for schema-less storage needs.
Caching & Message Broker	Redis Enterprise Cache for session state management; Apache Kafka or RabbitMQ for complex asynchronous events assignment engines.
Cloud Infrastructure	Amazon Web Services (AWS) or Google Cloud Platform (GCP). Managed containerization utilizing Kubernetes (EKS/GKE).
Map & Location APIs	Google Maps Platform APIs (Maps SDK, Places API, Distance Matrix API, Geocoding API).

3.2 Security, Compliance & Data Privacy

All data channels must operate over TLS 1.3 encryption in transit, with full AES-256 block encryption applied to resting user databases. The code architecture must be entirely immune to common OWASP Top 10 vulnerabilities. Compliance with localized data localization frameworks and payment standards (PCI-DSS) must be verified prior to testing phases.

4. DELIVERABLES, MILESTONES & MILESTONE PAYMENTS

The implementation lifecycle spans an estimated 24-week execution corridor. Payout structures are bound strictly to validated functional milestones.

Phase / Milestone	Key Verifiable Deliverables	Payment %
1. Inception & Design UI/UX	System Architecture Docs, Interactive Figma high-fidelity clickable prototype mockups covering all user roles.	15%

2. Core Backend Development	Database schema deployment, functional REST API endpoints, merchant configuration core module validation.	20%
3. Client App Beta Sync	Beta APK/TestFlight access for Customer & Driver modules executing live order cycles and webhook map polling.	25%
4. System Integration Testing	Full integration of SMS gateways, payment channels, and super-admin analytics consoles. End-to-end security audit logs.	20%
5. Launch & Deployment	Production app listing approval on Apple App Store & Google Play, live backend container production cluster handoff.	10%
6. Stabilization Support	Post-launch Hypercare phase execution for 60 calendar days, delivering final documented clean source code repositories.	10%

5. EVALUATION CRITERIA FOR PROPOSALS

Bidding software engineering firms will be appraised according to a combined Quality-Cost Based Selection (QCBS) framework: **70% Technical Merit** and **30% Financial Proposal competitiveness**.

- **Technical Competence (40%):** Direct evaluation of previous institutional experience launching production-grade geographic tracking systems and on-demand mobility applications.
- **Architecture & Engineering Design (20%):** Evaluation of proposed tech stack resilience, data safety frameworks, and scaling paradigms.
- **Personnel & Leadership Profiles (10%):** Background matrix of the assigned Solutions Architect, Lead Mobile Engineer, and Dedicated QA Lead.
- **Commercial Framework (30%):** Cost efficiency, transparency of resource hourly breakdown, and terms of post-production warranties.